THE KESTEN–STIGUM BOUND IS TIGHT FOR THE FOUR-STATE POTTS MODEL ON THE BINARY TREE

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Abstract. Consider the broadcast of a uniformly random colour from the root of the infinite binary tree, where each edge copies its parent's colour with probability $\lambda + (1-\lambda)/4$ and otherwise chooses one of the other three colours uniformly. We prove that for $0 < \lambda \le 1/\sqrt{2}$ the colours at depth n carry asymptotically no information about the root colour. Since reconstruction is possible for $2\lambda^2 > 1$ by the theorem of Kesten and Stigum, the Kesten–Stigum bound is the reconstruction threshold for the ferromagnetic four-state Potts channel on the binary tree. The proof is computer assisted. A certified upper bound on the second moment of the root posterior, obtained from an exact finite-state upgrade of the belief-propagation recursion, places the limiting posterior law in a small neighbourhood of the uniform law; an exact polynomial certificate then shows that no nontrivial belief-propagation fixed point lies in that neighbourhood.

1. Introduction

Let T be the infinite rooted binary tree. Fix $q \ge 2$ and $\lambda \in (0,1)$ and let $M_\lambda = \lambda I + (1-\lambda)J$ be the $q \times q$ Potts channel, where J has every entry $1/q$. The broadcast process assigns the root ρ a uniform colour $\sigma_\rho \in [q]$ and passes colours down each edge independently through M_λ. Write σ_{L_n} for the colours at depth n. The process is *reconstructible* if

$$\lim_{n\to\infty} \mathbb{E} \left\| \mathbb{P}(\sigma_\rho = \cdot \mid \sigma_{L_n}) - q^{-1}\mathbf{1} \right\|_{\mathrm{TV}} > 0,$$

and nonreconstructible otherwise. Kesten and Stigum [4] showed that reconstruction holds when $d\lambda^2 > 1$ on the d-ary tree, and for $q = 2$ this is sharp [1, 3]. For larger q the bound can fail to be sharp: Mossel [5] showed that for q large reconstruction holds below it, and Sly [8] proved that the bound is not tight for $q \ge 5$ and every large d, and that it is tight for $q = 3$ and large d. Mossel, Sly and Sohn [6] extended tightness to $q = 4$ for large d. Ricci-Tersenghi, Semerjian and Zdeborová [7] predicted, from an expansion of the cavity equations, that for the ferromagnetic four-state Potts model the Kesten–Stigum bound is tight for every degree distribution. For small degrees the rigorous results stop short of the bound; Bhatnagar and Maneva [2] developed a computational method for small trees and recorded the remaining gap.

The main result of this paper settles the binary tree at $q = 4$.

Theorem 1.1. *For the four-state Potts channel on the binary tree, the broadcast process is nonreconstructible for every $0 < \lambda \le 1/\sqrt{2}$. In particular the reconstruction threshold equals the Kesten–Stigum bound $\lambda = 1/\sqrt{2}$.*

The proof has two parts of different kinds. The first is an upper bound on the second moment $x_n = \mathbb{E}\sum_i (p_i - 1/4)^2$ of the root posterior $p = \mathbb{P}(\sigma_\rho = \cdot \mid \sigma_{L_n})$, valid for $n \ge 500$ (Theorem 3.1). It is obtained by running the belief-propagation recursion on an information upgrade of the posterior law that lives on a finite grid, in exact integer arithmetic. The second is a fixed-point exclusion. The limit of the posterior laws is a fixed point of the belief-propagation map (Theorem 4.1), and we construct an explicit function $K(a, b; m)$ of two messages and a

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scalar whose expectation at any fixed point equals $256(x - x_{\mathrm{out}})$ minus a positive multiple of m^3 (Theorem 6.1). A finite family of exact polynomial certificates shows $K \geq 0$ on the region permitted by the first part (Theorem 6.2). At a fixed point $x = x_{\mathrm{out}}$, so the second moment is zero.

Neither part alone suffices. The grid bound $x_n \leq 0.0453$ is far from zero, and an exclusion argument based on a local expansion around the uniform law covers a much smaller neighbourhood. The kernel K is chosen so that its region of validity reaches the grid bound. It combines the drift identity for the second moment with a family of 31 functions whose expectations vanish at every colour-symmetric law, and a sign condition on the third moment that holds at every fixed point (Theorem 5.1).

2. The recursion

Throughout $q = 4$. Let $\Delta_4 = \{p \in \mathbb{R}^4 : p_i \geq 0,\ \sum_i p_i = 1\}$. A posterior p is encoded by the *message* $a = \lambda(4p - \mathbf{1})$, which lies in $\lambda(4\Delta_4 - \mathbf{1})$ and has $\sum_i a_i = 0$. If the two children of a vertex send messages a and b, the posterior of the vertex is

$$P_i = \frac{(1 + a_i)(1 + b_i)}{\sum_j (1 + a_j)(1 + b_j)}, \qquad 4P - \mathbf{1} = \frac{H}{s},$$

where

$$c = a \cdot b, \qquad s = 1 + \tfrac{c}{4}, \qquad H_i = a_i + b_i + a_i b_i - \tfrac{c}{4}.$$

The outgoing message is $\lambda H/s$. Let μ be the law of the message at a vertex, taken unconditionally on the root colour. The law μ is invariant under colour permutations and has mean zero. The two children's observations are independent given the parent's colour but not unconditionally; their joint law has density $s(a, b)$ with respect to $\mu \otimes \mu$. We call $\mu \otimes \mu$ the base measure and write $\mathbb{E}$ for expectations with respect to it, so the belief-propagation map $\mathcal{T}_\lambda$ is

$$\int f \, d(\mathcal{T}_\lambda \mu) = \mathbb{E}\Big[s(a, b)\, f\big(\lambda H(a, b)/s(a, b)\big) \Big]. \tag{1}$$

We use the notation

$$A = |a|^2, \quad B = |b|^2, \quad t_a = \textstyle\sum_i a_i^3, \quad t_b = \sum_i b_i^3, \quad p_{ij} = \sum_k a_k^i b_k^j, \quad \ell = p_{21} + p_{12},$$

and the moments of μ,

$$m = \mathbb{E}A, \quad T = \mathbb{E}t_a, \quad V = \mathbb{E}A^2, \quad Q = \mathbb{E}\textstyle\sum_i a_i^4.$$

The second moment of the posterior is $x = \mathbb{E}\sum_i (p_i - 1/4)^2 = m/(16\lambda^2)$; at $\lambda = 1/\sqrt{2}$, $m = 8x$. From (1),

$$m_{\mathrm{out}} = \lambda^2 \mathbb{E}\Big[\frac{N}{s}\Big], \quad T_{\mathrm{out}} = \lambda^3 \mathbb{E}\Big[\frac{J}{s^2}\Big], \quad V_{\mathrm{out}} = \lambda^4 \mathbb{E}\Big[\frac{N^2}{s^3}\Big], \qquad N = \textstyle\sum_i H_i^2,\ J = \sum_i H_i^3. \tag{2}$$

3. The canonical bound

Let μ_n be the law of the root message given depth n. Its second moment is $16\lambda^2 x_n$.

Proposition 3.1. *Let*

$$U = \frac{1375940705703331}{30399297484750848} < 0.04527.$$

For every $0 < \lambda \leq 707107/10^6$ and every $n \geq 500$, $x_n \leq U$.

The proof compares experiments in the sense of Blackwell. A law of posteriors with uniform mean is an experiment about the root colour, and replacing it by a more informative experiment can only increase x, because x is the expectation of a convex function of the posterior. Belief propagation preserves the order: if each child's experiment is replaced by a more informative one, so is the parent's. A posterior p may be replaced by a random posterior with barycentre p (reveal additional information); mass may be moved to the point posterior of a colour chosen by a colour-equivariant rule (full revelation). The finite computation starts at full revelation, which is the depth-zero law, and alternates one exact belief-propagation step with two such upgrades.

Lemma 3.2 (symmetric lattice splitting)**.** *Let $N \geq 1$ and $p \in \Delta_4$. Write $Np_i = f_i + r_i$ with $f_i \in \mathbb{Z}$ and $0 \leq r_i < 1$, and let $R = \sum_i r_i \in \{0,1,2,3\}$. Define a law on $N^{-1}\mathbb{Z}^4 \cap \Delta_4$ as follows. If $R = 0$, the point mass at f/N. If $R = 1$, the point $(f + e_i)/N$ with weight r_i. If $R = 3$, the point $(f + \mathbf{1} - e_i)/N$ with weight $1 - r_i$. If $R = 2$, with $h = \min_i \min(r_i, 1 - r_i)$, the point $(f + e_i + e_j)/N$ with weight*

$$w_{ij} = \max(r_i + r_j - 1, 0) + h/3 \qquad (i < j).$$

In every case the weights are nonnegative, sum to one, have barycentre p, and depend equivariantly on p under colour permutations.

Proof. The cases $R \in \{0,1,3\}$ are immediate. Let $R = 2$ and sort the residuals $r_{(1)} \leq r_{(2)} \leq r_{(3)} \leq r_{(4)}$. Then $r_{(1)}+r_{(2)} \leq 1$, $r_{(1)}+r_{(3)} \leq 1$, $r_{(2)}+r_{(4)} \geq 1$ and $r_{(3)}+r_{(4)} \geq 1$. If $r_{(1)}+r_{(4)} \geq 1$, the positive excesses r_i+r_j-1 are those of the three pairs containing the largest residual, and for each coordinate i the excesses of the pairs containing i add up to $r_i-(1-r_{(4)})$. If $r_{(1)}+r_{(4)} \leq 1$, the positive excesses are those of the three pairs not containing the smallest residual, and the incident excesses add up to $r_i-r_{(1)}$. In both cases they add up to r_i-h, so adding $h/3$ to each of the three pairs at i gives total weight r_i at coordinate i. Summing over i counts each pair twice, so the weights add up to $R/2 = 1$. The barycentre is $(f+\sum_{i<j} w_{ij}(e_i+e_j))/N = (f+r)/N = p$. □

Proof of Theorem 3.1. Write $\lambda^* = 707107/10^6$; note $2(\lambda^*)^2 > 1$. Take $N = 36$ and work with laws supported on $\{v/36 : v \in \mathbb{Z}^4_{\geq 0},\ \sum v_i = 36\}$, grouped into the 478 orbits of the colour permutations. Starting from the point posterior, one step consists of the exact two-child update (1) at λ^* applied to the orbit law, the splitting of Theorem 3.2 applied to every resulting posterior, and grouping of equal posteriors. The orbit masses are then rounded down to integer multiples of 2^{-48} and the deficit is placed on the orbit of full revelation. All transition probabilities have the common denominator $288(36 \cdot 10^6)^2$, and every quantity is an exact integer.

By Theorem 3.2, splitting is an upgrade. Rounding down and moving the deficit to full revelation is an upgrade because each orbit is colour-symmetric with uniform mean. By induction the computed law at step n is more informative than μ_n at λ^*. After 500 steps its second moment equals U exactly. Since x_n is nonincreasing in n ($\sigma_{L_{n+1}}$ is obtained from σ_{L_n} by independent randomness, so it is a garbling of σ_{L_n}), the bound holds for all $n \geq 500$. For $\lambda < \lambda^*$ it follows from Theorem 3.3. □

Lemma 3.3. *For $0 < \lambda' < \lambda < 1$ and every n, $x_n(\lambda') \leq x_n(\lambda)$.*

Proof. Put $\theta = \lambda'/\lambda$. On each edge draw independently a permutation R_v of the colours, equal to the identity with probability θ and uniform otherwise, and set $G_\rho = \mathrm{id}$, $G_v = G_{\mathrm{parent}(v)} \circ R_v$. If σ is the λ-broadcast, then $\sigma'_v = G_v(\sigma_v)$ is the λ'-broadcast, because averaging over R_v gives the transition $M_\lambda M_\theta = M_{\lambda'}$, which does not depend on $G_{\mathrm{parent}(v)}$. The root colour is unchanged and σ'_{L_n} is a function of σ_{L_n} and independent randomness, so the λ' observation is a garbling of the λ observation and Jensen's inequality gives the claim. □

4. The limit law

Lemma 4.1. *Fix $\lambda \in (0,1)$. The laws μ_n converge weakly to a colour-symmetric law μ_∞ with mean zero, $\mu_\infty = \mathcal{T}_\lambda \mu_\infty$, and $16\lambda^2 x_n \to \int |a|^2 \, d\mu_\infty$. The process is nonreconstructible if and only if μ_∞ is the point mass at 0.*

Proof. Let $\mathcal{F}_n$ be generated by all colours at depth at least n. By the Markov property $\mathbb{P}(\sigma_\rho = i \mid \mathcal{F}_n) = \mathbb{P}(\sigma_\rho = i \mid \sigma_{L_n})$, and the σ-fields decrease, so Lévy's downward theorem gives almost sure and L^1 convergence of the posteriors. Hence μ_n converges weakly and the second moments converge. Symmetry and the mean pass to the limit. In (1) the density s is at least $1 - \lambda^2 > 0$ and the integrand is a bounded continuous function of (a, b), so $\mathcal{T}_\lambda$ is weakly continuous, and $\mu_{n+1} = \mathcal{T}_\lambda \mu_n$ gives the fixed-point equation. The last statement follows from $\mathbb{E}\|p - \frac{1}{4}\mathbf{1}\|_{\mathrm{TV}} \le \sqrt{x_n}$. □

Lemma 4.2 (support)**.** *Let $\varrho = 677/1000$ and $0 < \lambda \le 1/\sqrt{2}$. Every fixed point μ of $\mathcal{T}_\lambda$ is supported in $\varrho(4\Delta_4 - \mathbf{1})$.*

Proof. A zero-sum vector lies in $r(4\Delta_4 - \mathbf{1})$ exactly when its coordinates are at least $-r$. Suppose $a, b \in r(4\Delta_4 - \mathbf{1})$ with $0 \le r < 1$. For fixed b the coordinate P_i is a linear-fractional function of a with positive denominator, so its minimum over the simplex is attained at a vertex, and likewise in b. At a pair of vertices the smallest coordinate of P is $(1-r)^2/(4(1+3r^2))$ when the two favoured colours agree and $(1-r)^2/(4(1-r^2))$ otherwise. Hence the outgoing message has coordinates at least $-f(r)$ with $f(r) = 2\ell r(1+r)/(1+3r^2)$, $\ell = 707107/10^6$. The function f is increasing on $[0,1]$. Every message lies in $\ell(4\Delta_4 - \mathbf{1})$, and since $\mu = \mathcal{T}_\lambda^k \mu$ for every k, μ is supported in $r_k(4\Delta_4 - \mathbf{1})$ where $r_0 = \ell$ and $r_{k+1} = f(r_k)$. The sequence decreases to the fixed point $0.6757\ldots$ of f, and an exact rational computation shows $r_k \le \varrho$ after finitely many steps. □

5. The third moment at a fixed point

Proposition 5.1. *Let $0 < \lambda \le 1/\sqrt{2}$ and let μ be a colour-symmetric fixed point of $\mathcal{T}_\lambda$ with second moment m and third moment T. Then*

$$\tfrac{69763}{500000}\, m^2 \le \left(\lambda^{-3} - 2 + \tfrac{m}{6}\right) T \le \tfrac{7}{6}\, m^2.$$

In particular $T \ge 0$.

Proof. For independent colour-symmetric a, b supported in $\ell(4\Delta_4 - \mathbf{1})$ we show

$$\tfrac{69763}{500000}\, m_a m_b \le \mathbb{E}\Big[\frac{J}{s^2}\Big] - \big(1 - \tfrac{m_b}{12}\big) T_a - \big(1 - \tfrac{m_a}{12}\big) T_b \le \tfrac{7}{6}\, m_a m_b. \tag{3}$$

With $a = b$ in law and $T_{\mathrm{out}} = T$, (2) turns (3) into the statement, and $\lambda^{-3} \ge 2\sqrt{2} > 2$ gives $T \ge 0$.

For a coset C of the Klein subgroup $\{e, (01)(23), (02)(13), (03)(12)\}$ in S_4 write b_π for b with coordinates permuted by π, and put

$$G_C = \frac{1}{|C|} \sum_{\pi \in C} \frac{J(a, b_\pi)}{s(a, b_\pi)^2} - t_a - t_b + \frac{B t_a + A t_b}{12}, \qquad R_C = \frac{3}{|C|} \sum_{\pi \in C} (a \cdot b_\pi)^2 - AB.$$

For each of the six cosets we prove the pointwise inequality

$$\beta\, AB \le G_C + R_C/2 \le M_C\, AB, \qquad \beta = 69763/500000,$$

with constants M_C whose average is $3498859/3000000 < 7/6$. Sort a and parametrise both messages by barycentric coordinates $z, w \ge 0$ on the ordered simplex with vertices

$$\ell(-1,-1,-1,3), \quad \ell(-1,-1,1,1), \quad \ell(-1, \tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}), \quad 0.$$

After clearing the positive denominators $s_\pi \geq 1-\ell^2$ and the homogenising factors, $G_C + R_C/2$ and AB become polynomials of bidegree $(11,11)$ in (z,w) with the same 129600 monomials, and the two inequalities hold coefficient by coefficient. The six cosets have the same size, so the average of $G_C + R_C/2$ over them lies between βAB and $\big(\frac{1}{6}\sum_C M_C\big)AB \leq \frac{7}{6}AB$, and it is the average over all of S_4. Averaging over cosets, using colour symmetry of each law separately, and taking expectations gives (3), since the average of R_C over all cosets is $3(a\cdot b)^2$ averaged over S_4 minus AB, which is zero for zero-sum vectors. □

6. The kernel

6.1. **Moment identities.** Let a, b be independent with the same colour-symmetric law of mean zero. For every scalar function w invariant under colour permutations,

$$\mathbb{E}[w(a)a_i] = 0, \qquad \mathbb{E}[w(a)a_ia_j] = \tfrac{1}{3}\mathbb{E}[w(a)A]\big(\delta_{ij} - \tfrac{1}{4}\big), \tag{4}$$

and the third-moment tensor is $\mathbb{E}a_i^3 = T/4$, $\mathbb{E}a_i^2a_j = -T/12$, $\mathbb{E}a_ia_ja_k = T/12$ for distinct indices. Conditioning on one of the two messages and applying these gives the expectations in Table 1.

expression	expectation	expression	expectation
c^2	$m^2/3$	p_{33}	$T^2/4$
$(A+B)c^2$	$2mV/3$	$p_{42}+p_{24}$	$mQ/2$
$c\,\ell$	$2mT/3$	$c(p_{31}+p_{13})$	$2mQ/3$
c^3	$2T^2/3$	$p_{21}^2+p_{12}^2$	$\frac{2m}{3}(Q-V/4)$
p_{22}	$m^2/4$	$p_{21}p_{12}$	$T^2/3$
$p_{32}+p_{23}$	$mT/2$	$(A+B)p_{22}$	$mV/2$
$c\,p_{22}$	$T^2/3$		

Table 1. Expectations under the base measure.

Expanding $N = A+B+2c+2\ell+p_{22}-c^2/4$ and $1/s = 1-c/4+c^2/(16s)$, the table gives $\mathbb{E}N = 2m+m^2/6$ and $\mathbb{E}[Nc] = 2m^2/3+4mT/3+T^2/6$. At $\lambda = 1/\sqrt{2}$, (2) then yields

$$256\,(x - x_{\mathrm{out}}) = \tfrac{16}{3}mT + \tfrac{2}{3}T^2 - \mathbb{E}\Big[\frac{Nc^2}{s}\Big]. \tag{5}$$

6.2. **Definition of the kernel.** For messages a, b and a scalar $m \geq 0$ let

$$K(a,b;m) = \tfrac{2}{3}t_at_b - \frac{Nc^2}{s} - \beta\, m\, AB + \sum_{j=0}^{32} g_j\,\phi_j(a,b;m), \qquad \beta = 10^{-6},$$

where the functions ϕ_j are listed in Table 2 and the rational coefficients g_j in Section A. Here $F_3 = J/s^2$, $F_4 = N^2/s^3$, $s_a = \sum_i a_i^4$, $s_b = \sum_i b_i^4$, p_{31} stands for $\sum_k(a_k^3b_k + a_kb_k^3)$, and $\kappa = c\,\ell - (Bt_a + At_b)/3$. The coefficients of ϕ_{25}, ϕ_{26}, ϕ_{27} and ϕ_{28} are zero, so K is affine in m.

Proposition 6.1. *Let μ be a colour-symmetric fixed point of $\mathcal{T}_{1/\sqrt{2}}$ with second moment m and third moment T, and let a, b be independent with law μ. Then*

$$\mathbb{E}\,K(a,b;m) = 256\,(x - x_{\mathrm{out}}) - \beta m^3 - g^*mT, \qquad g^* = \tfrac{16}{3} - g_0(2\sqrt{2}-2) > \tfrac{863}{375000000}.$$

Proof. At $\lambda = 1/\sqrt{2}$ the fixed-point equations (2) read $\mathbb{E}F_3 = 2\sqrt{2}\,T$ and $\mathbb{E}F_4 = 4V$. Hence $\mathbb{E}\phi_0 = m(2\sqrt{2}-2)T$ and $\mathbb{E}\phi_1 = 2mV - m(4V-2V) = 0$. The expectations of $\phi_2, \dots, \phi_{32}$ vanish: for $\phi_5, \phi_6, \phi_7, \phi_9, \phi_{10}, \phi_{11}, \phi_{12}, \phi_{14}, \phi_{15}, \phi_{16}, \phi_{22}$ by (4), for ϕ_{23} by the symmetry of the law of b, for ϕ_{24} through ϕ_{28} by independence and $\mathbb{E}A = \mathbb{E}B = m$, and for the others by Table 1. Finally

j	ϕ_j	j	ϕ_j
0	$m(F_3 - t_a - t_b)$	17	$m\kappa$
1	$AB(A+B) - m(F_4 - A^2 - B^2)$	18	$m\big(p_{32} + p_{23} - (Bt_a + At_b)/4\big)$
2	$(A+B)(c^2 - AB/3)$	19	$c\,p_{22} - t_a t_b/3$
3	$m(c^2 - AB/3)$	20	$p_{33} - t_a t_b/4$
4	κ	21	$p_{42} + p_{24} - (Bs_a + As_b)/4$
5	$c(A+B)$	22	$mc(A^2 + B^2)$
6	$c(A^2 + B^2)$	23	$24c^4 - \sum_{\pi \in S_4} (a \cdot b_\pi)^4$
7	$Bp_{21} + Ap_{12}$	24	$AB(A+B) - m(A^2 + B^2)$
8	$c^3 - \frac{2}{3} t_a t_b$	25	$Bt_a + At_b - m(t_a + t_b)$
9	cAB	26	$m(A-m)(B-m)$
10	$mc(A+B)$	27	$(A+B)(A-m)(B-m)$
11	$c\,t_a t_b$	28	$m^2(A+B-2m)$
12	$m\ell$	29	$c\,p_{31} - (Bs_a + As_b)/3$
13	$m(p_{22} - AB/4)$	30	$p_{21}^2 + p_{12}^2 - \big(Bs_a + As_b - AB(A+B)/4\big)/3$
14	$m\,p_{31}$	31	$p_{21}p_{12} - t_a t_b/3$
15	$mc(t_a + t_b)$	32	$(A+B)(p_{22} - AB/4)$
16	$m\ell(A+B)$		

TABLE 2. The functions in the kernel.

$\mathbb{E}[\frac{2}{3} t_a t_b] = \frac{2}{3} T^2$ and $\mathbb{E}[\beta m AB] = \beta m^3$; combining with (5) gives the formula. The bound on g^* uses $g_0 = 32189/5000$ and $\sqrt{2} < 70711/50000$. □

Proposition 6.2. *Let $\varrho = 677/1000$ and $M = 8U = 1375940705703331/3799912185593856$. Then $K(a, b; m) \geq 0$ for all $a, b \in \varrho(4\Delta_4 - \mathbf{1})$ and $0 \leq m \leq M$.*

Proof. Since K is affine in m, it suffices to treat $m = 0$ and $m = M$. Every function in K is invariant under a simultaneous permutation of the colours of a and b, so we may assume $a_0 \leq a_1 \leq a_2 \leq a_3$; the coordinates of b are then sorted by one of the 24 permutations π. Write a and the sorted b in barycentric coordinates $z, w \geq 0$ of the ordered simplex with vertices $\varrho(-1,-1,-1,3)$, $\varrho(-1,-1,1,1)$, $\varrho(-1, \frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ and 0, and split each of the two charts into the cells $z_0 \geq z_3$ and $z_3 \geq z_0$ (respectively for w), substituting $z_0 = z_0' + z_3$ or $z_3 = z_3' + z_0$. This gives $24 \times 4 \times 2 = 192$ cases. In each case

$$G = K \cdot P^5 Q^5 S^3 (UV)^e, \qquad U = \sum z_i,\ V = \sum w_i,\ P = 3 \cdot 10^6 U,\ Q = 3 \cdot 10^6 V,$$

where $a_i = \alpha_i / P$ and $b_i = \alpha_i' / Q$ with integer linear forms α_i, α_i' in the cell variables and $S = 4PQ + \sum_i \alpha_i \alpha_i'$ is the cleared form of s, is a polynomial in the cell variables, and $S > 0$ because $s \geq 1 - \varrho^2$. Let

$$\Sigma_1 = AB(A+B) - 2t_a t_b,\ \Sigma_2 = (A+B)(AB - c^2),\ \Sigma_3 = m(AB - c^2),\ \Sigma_4 = AB(A+B) + 2t_a t_b.$$

Each Σ_k is nonnegative: Σ_2, Σ_3 by Cauchy–Schwarz, and Σ_1, Σ_4 because $|t_a t_b| \leq (AB)^{3/2} \leq AB(A+B)/2$. For each case there are rationals $\gamma_k \geq 0$ such that every coefficient of

$$G - \sum_k \gamma_k \, \Sigma_k \, P^5 Q^5 S^3 (UV)^e$$

is nonnegative; the degree e is 1 in 176 cases and 2 in 16, $\gamma = 0$ in 168 cases and $\gamma = (1/3, 0, 0, 0)$ in the remaining 24. Since the cell variables are nonnegative and the factors P, Q, S, U, V are positive, $K \geq \sum_k \gamma_k \Sigma_k \geq 0$ on every cell. □

7. Proof of the main theorem

Let $\lambda = 1/\sqrt{2}$ and let μ_∞ be the limit law of Theorem 4.1, with moments m and T and posterior second moment x. By Theorems 3.1 and 4.1, $x \le U$, so $m = 8x \le M$. By Theorem 4.2 the law is supported in $\varrho(4\Delta_4 - \mathbf{1})$, so Theorem 6.2 applies to every pair in its support and $\mathbb{E}K \ge 0$. At a fixed point $x_{\mathrm{out}} = x$, so Theorem 6.1 gives $\beta m^3 + g^* m T \le 0$. By Theorem 5.1, $T \ge 0$, and therefore $m = 0$. Thus μ_∞ is the point mass at zero and the process is nonreconstructible at $\lambda = 1/\sqrt{2}$. For $\lambda < 1/\sqrt{2}$ the same follows from Theorem 3.3. Reconstruction for $\lambda > 1/\sqrt{2}$ is the Kesten–Stigum theorem. □

Remark 7.1. Since $\mathbb{E}\phi_{24} = 0$ at every colour-symmetric law, the coefficient g_{24} does not affect Theorem 6.1; its value only serves the certificates of Theorem 6.2. The certificates are not slack: with $\gamma = 0$ in the cases where $\gamma \ne 0$, or with the shift $1743/480965$ in g_{24} replaced by $-1/10$, some coefficients become negative.

8. The computations

All computations use exact integer or rational arithmetic; no floating-point value enters a bound. The grid computation of Theorem 3.1 handles the 114481 unordered pairs of the 478 orbits per step and stores masses as integers over 2^{48}. Two independent implementations of Theorem 3.2 and (1) agree on all 500 intermediate moments and on the final value U. The coefficient comparisons of Theorem 5.1 cover 6×129600 inequalities. The 192 certificates of Theorem 6.2 were computed with `python-flint`; each polynomial is rebuilt from the definitions of Table 2, and its value was compared with a direct evaluation of K at rational points. The certificates, the coefficient files and the programs that rebuild and check them are at `https://github.com/yspennstate/potts-four-state-binary-tree`.

Appendix A. The coefficients

j	g_j	j	g_j
0	32189/5000	17	8086351/894289
1	307513/304635	18	314989/778090
2	1349531/958408	19	−293112/998719
3	13119169/971828	20	390816/998719
4–7	0	21	2119698/387547
8	2046290/998719	22	−760908/859123
9	0	23	−9410/911487
10	10803745609/1218540000	24	g_{24} (below)
11	649352/585675	25–28	0
12	−96567/5000	29	−1454253/671354
13	−33529637/939631	30	−1640643/978656
14	−96567/5000	31	−146556/998719
15	1007541/284861	32	−505057/629438
16	2858842/731359		

$$g_{24} = -\frac{4467359758744941897642393419111 3}{1462309266444848905334068011200160} + \frac{1743}{480965}$$
$$= -\frac{3787526362446847631199691392941357033}{1406639152671293507508000042013769908 80} \approx -0.026926.$$

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [9] and working together inside the Brain framework [11], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [10], the Math Brain repository [9] and the Brain framework repository [11].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Code and data availability

The code and data for this paper are at https://github.com/yspennstate/potts-four-state-binary-tree. The paper is part of the Math Brain project, whose public repository is https://github.com/yspennstate/math-brain-public [9].

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. In October 2026 OpenAI GPT models, running as research tasks of that project, wrote the proofs and the programs that produce the computer-assisted steps. Claude models (Anthropic), working in Claude Code, audited every piece with programs of their own, wrote the kernel certificates of Theorem 6.2, assembled the argument, and wrote the paper. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

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